An affine Singer layer on $\mathbb{F}_2^6$, its (T, D, K) triple, and $\langle T, D\rangle = \mathrm{AGL}(1, 64)$

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Abstract

Let $V = \mathbb{F}_2^6$ carry the ordered basis $G = (63, 33, 18, 49, 1, 21)$, and define $T(x) = 63 \oplus \mathrm{coords}_G(x)$, where the coordinates are read as a six-bit word with the first coordinate most significant. We prove that T has fixed point 38 and one 63-cycle; its linear part has primitive characteristic polynomial $x^6 + x^5 + x^3 + x^2 + 1$. The maps $D = (\mathrm{id} \oplus T)^{-1}$ and $K = (\mathrm{id} \oplus D)^{-1}$ have cycle types $1 + 63$ and $1 + 7 \cdot 9$, with fixed points 41 and 63, respectively, and satisfy $(\mathrm{id} \oplus K)^{-1} = T$. We give an elementary proof that $\langle T, D\rangle = \mathrm{AGL}(1, 64)$, of order 4032. For every $k \geq 1$, we construct an analogous triple on $\mathbb{F}_{2^{2k}}$ with T, D of type $1 + (2^{2k} - 1)$ and K of type $1 + (2^k - 1) \times (2^k + 1)$, generating $\mathrm{AGL}(1, 2^{2k})$. The linear part of K has characteristic polynomial an irreducible degree-$2k$ factor of Φ_{2^k+1}. The existence proof uses the classical prescribed-trace primitive-element theorem, reconstructed in Appendix B with a deterministic certificate for $k \leq 29$; no priority is claimed for this input. Explicit coordinate bases are verified for $k = 1, 2, 3, 4$, with group orders $12, 240, 4032, 65280$. We also classify the orbits of $F(f) = (\mathrm{id} \oplus f)^{-1}$ on scalar affine maps over $\mathbb{F}_{2^{2k}}$. The note brings together this coordinate realization, the triple identities, and the affine group-generation argument.

1 Introduction

The 64 words of $\mathbb{F}_2^6$ support a classical combinatorial structure with operations of complement, reversal, and a recombination that reads a word through its coordinates in a fixed generator basis. This note isolates one such operation — the element transformation T — and shows that it is exactly an affine Singer map (layer) — that is, an affine map whose linear part is a Singer cycle. The associated triple (T, D, K) and the group $\langle T, D\rangle$ then reduce to elementary finite-field statements, with a uniform analogue for $2k$-bit systems (an existence theorem for all k, §6 and Appendix B; the concrete coordinate presentation is verified for $k = 1, 2, 3, 4$). Related families of permutations over $\mathbb{F}_2^n$ with prescribed cycle structure and algebraic degree were studied by Gravel, Panario and Thomson [7]. The underlying even-characteristic existence statement is classical (see [4] and [5, Theorem 1, p. 1]; an equivalent square-order formulation is [3, Theorem 1.5, p. 170]); Cohen and Presern [6] subsequently gave a self-contained account of the prescribed-trace theorem with reduced computation. Appendix B reconstructs the even-characteristic trace-one case and provides an explicit finite certificate; neither the existence theorem nor its character-sum method is claimed as new.

The focus of this note is the explicit coordinate realization in dimension six and its organization into a triple linked to affine group generation. We give the concrete basis, fixed points, characteristic polynomials, and cycle types; connect the three-step operator identity to a direct group-generation proof; and extend this presentation to every even dimension using the classical existence input.

Proposition 7.1 records the elementary orbit classification of this operator on scalar affine maps. The finite certificates make the existence proof reproducible, and the Lean formalization covers the group-generation argument, with its scope stated in Appendix A. These contributions are an explicit realization and a unified presentation of the affine structure. The classical names of the words are not used.

2 Notation

$V = \mathbb{F}_2^6$; coordinates $x = (x_1, \ldots, x_6)$ with x_1 the most significant bit (value 32) and x_6 the least (value 1); a word is identified with its integer value in $\{0, \ldots, 63\}$. Basis $G = (\mathtt{63}, \mathtt{33}, \mathtt{18}, \mathtt{49}, \mathtt{1}, \mathtt{21})$ (a fixed ordered basis — the six words are linearly independent over $\mathbb{F}_2$, by Gaussian elimination; non-uniqueness is discussed in §6); write g_j for the j-th entry, $j = 0, \ldots, 5$. $\mathrm{coords}_G(x) = (c_1, \ldots, c_6)$ is the unique vector with $x = \bigoplus_{i=1}^{6} c_i\, g_{i-1}$, read as the six-bit word $c_1 \cdots c_6$ (c_1 most significant); $\oplus$ is bitwise XOR, i.e. addition in $\mathbb{F}_2^n$.

3 T is an affine Singer map

3.1 Definition and basic checks

$T(x) = 63 \oplus \mathrm{coords}_G(x)$. Verified identities: $T(0) = 63$, $T(41) = 0$, $T(38) = 38$; for example $\mathrm{coords}_G(38) = (0, 1, 1, 0, 0, 1)$, read as $011001 = 25$, so $T(38) = 63 \oplus 25 = 38$, and $\mathrm{coords}_G(41) = (1, 1, 1, 1, 1, 1) = 63$, so $T(41) = 63 \oplus 63 = 0$.

Lemma 3.1. *T is affine: $T(x) = L_T x + b$ with $b = T(0) = 63$ and $L_T(x) = T(x) \oplus 63$.*

Lemma 3.2. $\mathrm{Fix}(T) = \{38\}$*; on the remaining* 63 *points T is a single cycle (type* $1 + 63$*, order* 63*), equivalently L_T has order* 63 *(Proposition 3.3).*

Proposition 3.3. *Identify $(V, +)$ with $\mathbb{F}_{64}$; then L_T is multiplication by a primitive element of $\mathbb{F}_{64}$, a Singer cycle [1]. The characteristic polynomial of L_T is*

$$p_T(t) = t^6 + t^5 + t^3 + t^2 + 1, \tag{1}$$

computed directly as $\det(tI + L_T)$ *(code appendix), primitive of degree* 6*.*

Proof. The matrix in Appendix A gives the displayed determinant. Modulo p_T, repeated squaring and multiplication give

$$t^{64} = t, \qquad \gcd(p_T, t^8 + t) = \gcd(p_T, t^4 + t) = 1.$$

The degree-6 irreducibility criterion therefore shows that p_T is irreducible. In $\mathbb{F}_2[t]/(p_T)$ the residue $\bar{t}$ satisfies

$$\bar{t}^{63} = 1, \qquad \bar{t}^{21} = \bar{t}^4 + \bar{t}^2 + \bar{t} + 1 \neq 1, \qquad \bar{t}^9 = \bar{t}^2 + \bar{t} \neq 1.$$

Since $63 = 3^2 \cdot 7$, its order is 63. The irreducible characteristic polynomial identifies L_T with multiplication by $\bar{t}$ on this field. Finally, $T(38) = 38$ implies $T(x) \oplus 38 = L_T(x \oplus 38)$, proving the fixed-point and cycle claims of Lemma 3.2. □

3.2 Powers and $63 = 7 \cdot 9 = 3 \cdot 21$

Since L_T has order 63, T^d for $d \mid 63$ has type $1 + d \times (63/d)$ (one fixed point plus d cycles of length $63/d$): $T^1 = 1 + 63$, $T^3 = 1 + 3 \cdot 21$, $T^7 = 1 + 7 \cdot 9$, $T^9 = 1 + 9 \cdot 7$, $T^{21} = 1 + 21 \cdot 3$ (verified).

4 The (T, D, K) triple and the fixed-point operator

For a permutation f of V such that $\mathrm{id} \oplus f$ is again a permutation, define the *fixed-point operator* $F(f) = (\mathrm{id} \oplus f)^{-1}$, where $(\mathrm{id} \oplus f)(x) = x \oplus f(x)$. By Proposition 3.3, identify the linear part of T with multiplication by a primitive $\alpha \neq 1$. Then $\mathrm{id} \oplus T$ has nonzero slope $1 \oplus \alpha$ and is therefore a permutation. Its inverse $D = F(T)$ has slope $(1 \oplus \alpha)^{-1}$, and $1 \oplus (1 \oplus \alpha)^{-1} = \alpha/(1 \oplus \alpha) \neq 0$, so $\mathrm{id} \oplus D$ is also a permutation. Thus $D = F(T) = (\mathrm{id} \oplus T)^{-1}$ and $K = F(D) = (\mathrm{id} \oplus D)^{-1}$ are defined; since $F^3(T) = T$ below, $\{T, D, K\}$ is the F-orbit of T. Verified facts:

- cycle types $T = D = 1 + 63$, $K = 1 + 7 \cdot 9$ (seven 9-cycles plus one fixed point);
- fixed points $\mathrm{Fix}(T) = \{38\}$, $\mathrm{Fix}(D) = \{41\}$, $\mathrm{Fix}(K) = \{63\}$;
- $F(K) = T$, i.e. $(\mathrm{id} \oplus K)^{-1} = T$; hence $F^3 = \mathrm{id}$ with $T \to D \to K \to T$;
- offset-fixed three-cycle: $D(0) = 38 = \mathrm{Fix}(T)$, $K(0) = 41 = \mathrm{Fix}(D)$, $T(0) = 63 = \mathrm{Fix}(K)$;
- linear parts: since $L_D = (I + L_T)^{-1}$, its characteristic polynomial is the reciprocal of (1), namely $p_D(t) = t^6 p_T(1/t) = t^6 + t^4 + t^3 + t + 1$ (primitive). Separately, $p_T(t + 1) = p_T(t)$ in characteristic two (directly, $(t+1)^6+(t+1)^5+(t+1)^3+(t+1)^2+1$ reduces to $t^6+t^5+t^3+t^2+1$); L_K has $\Phi_9(t) = t^6 + t^3 + 1$ (cyclotomic, non-primitive, order 9).

Lemma 4.1. *For $g \in \{T, D, K\}$, the unique fixed point of g is $F(g)(0)$, i.e.* $\mathrm{Fix}(g) = \{F(g)(0)\}$.

Lemma 4.2. *In characteristic two, the fractional linear map $z \mapsto 1/(1 + z)$ has order three: $z \mapsto 1/(1+z) \mapsto 1+1/z \mapsto z$ (where defined), and for the operator $F(f) = (\mathrm{id} \oplus f)^{-1}$ on affine maps $f(x) = \alpha x \oplus c$ with $\alpha, 1 \oplus \alpha \neq 0$, one has $F^3(f) = f$; the slopes cycle as $\alpha \mapsto (1 \oplus \alpha)^{-1} \mapsto 1 \oplus \alpha^{-1} \mapsto \alpha$.*

Proof. Set $\beta = (1 \oplus \alpha)^{-1}$ and $\eta = 1 \oplus \alpha^{-1}$. Then $F(f)(x) = \beta x \oplus \beta c$, $F^2(f)(x) = \eta x \oplus \alpha^{-1} c$, and $F^3(f)(x) = \alpha x \oplus c = f(x)$. In particular, $\alpha\beta\eta = 1$. □

Remark. The fractional-linear statement holds in characteristic two and fails in odd characteristic; the affine identity $F^3 = \mathrm{id}$ is genuinely characteristic-two: over a field of odd characteristic, whenever the iterates are defined (equivalently, $(1 + \alpha)(2 + \alpha)(3 + 2\alpha) \neq 0$), $F^3(f)(x) = \frac{\alpha+2}{2\alpha+3} x - \frac{c}{2\alpha+3} \neq \alpha x + c = f(x)$ whenever $c \neq 0$. Thus the $\mathbb{F}_2$ setting is essential for the third-order affine cycle, not merely a convenience.

The identity is affine-general: by Lemma 4.2, in field coordinates $T(x) = \alpha x \oplus c$, so $F(T)(0) = D(0) = (1 \oplus \alpha)^{-1} c$ solves $x = \alpha x \oplus c$, i.e. $D(0) = \mathrm{Fix}(T)$, and the slope cycle $\alpha \mapsto (1 \oplus \alpha)^{-1} \mapsto 1 \oplus \alpha^{-1} \mapsto \alpha$ gives the two remaining identities. For the six-bit layer the values are $F(T)(0) = 38$, $F(D)(0) = 41$, $F(K)(0) = 63$. Conceptually, T is the launching map — an affine Singer layer read through the fixed generator basis (a change of basis) — while the fixed-point operator F keeps the third-order cycle running: $T \mapsto D \mapsto K \mapsto T$, carrying the fixed point $38 \mapsto 41 \mapsto 63 \mapsto 38$.

5 The group $\langle T, D \rangle$

Theorem 5.1. $\langle T, D \rangle = \mathrm{AGL}(1, 64)$*; its order is* $4032 = 64 \cdot 63$.

Proof. By Proposition 3.3, choose the field identification in which the linear part of T is multiplication by a primitive α (order 63); then $T(x) = \alpha x \oplus c$ with $c = T(0) = 63 \neq 0$. Since $(\mathrm{id} \oplus T)(x) = (1 \oplus \alpha)x \oplus c$, the inverse is affine, $D(x) = \beta x \oplus \beta c$ with $\beta = (1 \oplus \alpha)^{-1}$. Every word of $\langle T, D \rangle$ is therefore affine with multiplier in $\mathbb{F}_{64}^* = \langle \alpha \rangle$, so $\langle T, D \rangle \subseteq \mathrm{AGL}(1, 64)$. For

the converse, $T \circ D$ and $D \circ T$ have the same multiplier $\alpha\beta$: $T \circ D(x) = \alpha\beta x \oplus \alpha\beta c \oplus c$ and $D \circ T(x) = \alpha\beta x$, so $T \circ D(x) = D \circ T(x) \oplus c\beta$ (as $\alpha\beta \oplus 1 = \beta$), with $c\beta = c/(1 \oplus \alpha) \neq 0$; hence the commutator $[T, D] = TDT^{-1}D^{-1}$ has multiplier 1 and is the nonzero translation $x \mapsto x \oplus c/(1 \oplus \alpha)$. Conjugating it by T^j, $j = 0, \ldots, 62$, gives the translations $x \mapsto x \oplus \alpha^j c/(1 \oplus \alpha)$, which, as α is primitive, are all nonzero translations together with the identity. Since the powers of T supply every multiplier $\mathbb{F}_{64}^* = \langle\alpha\rangle$, together with these translations $\langle T, D\rangle$ contains $\{x \mapsto ax \oplus b : a \in \mathbb{F}_{64}^*, b \in \mathbb{F}_{64}\} = \mathrm{AGL}(1, 64)$ (cf. [2]). □

6 Uniform statement for $2k$-bit words

Let $V_{2k} = \mathbb{F}_{2^{2k}}$; write $M = 2^{2k} - 1 = (2^k - 1)(2^k + 1)$. In field coordinates an affine Singer layer is a map $T(x) = \alpha x \oplus b$ whose linear part is multiplication by a primitive element α (order M).

Lemma 6.1. *For every $k \geq 1$, $\mathrm{ord}_{2^k+1}(2) = 2k$.*

Proof. Let $m = \mathrm{ord}_{2^k+1}(2)$. Since $2^k \equiv -1 \pmod{2^k + 1}$, we have $m \mid 2k$, while $m \nmid k$. Suppose, for contradiction, that $m < 2k$. Since $m \mid 2k$ but $m \nmid k$, m is even; write $m = 2d$ with $d \mid k$, and $m < 2k$ gives $d < k$. The subgroup $\langle 2\rangle \subset (\mathbb{Z}/(2^k+1)\mathbb{Z})^\times$ is cyclic of order $2d$, so its unique element of order 2 is 2^d. But $2^k \equiv -1$ also has order 2 in this subgroup; hence $2^d \equiv -1 \pmod{2^k+1}$. Thus $2^k + 1 \mid 2^d + 1$, contradicting $0 < 2^d + 1 < 2^k + 1$. Therefore $m = 2k$. □

For $q = 2^k$, define

$$\mathrm{PTE}(k): \quad \exists \beta \in \mathbb{F}_{q^2}^*, \quad \mathrm{ord}(\beta) = q^2 - 1, \quad \mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\beta) = 1.$$

The following theorem supplies the required existence input; its proof is given in Appendix B.

Theorem 6.2 (PTE existence)**.** *For every $k \geq 1$, $\mathrm{PTE}(k)$ holds.*

Proof. See Appendix B. □

Theorem 6.3. *For every $k \geq 1$ there is an affine Singer layer T (with nonzero translation) on $\mathbb{F}_{2^{2k}}$ for which $D = (\mathrm{id} \oplus T)^{-1}$ and $K = (\mathrm{id} \oplus D)^{-1}$ are defined, T and D have cycle type $1 + M$, K has cycle type $1 + (2^k - 1) \times (2^k + 1)$, and the fixed-point operator satisfies $F^3 = \mathrm{id}$ with $T \mapsto D \mapsto K \mapsto T$.*

Proof. Take $T(x) = \alpha x \oplus b$ with $b \neq 0$. The linear parts of T, D, K are α, $(1 \oplus \alpha)^{-1}$, and $\kappa = 1 \oplus \alpha^{-1} = (1 \oplus \alpha)/\alpha$ respectively (independent of the translation b); the three slopes are cycled by the map $z \mapsto 1/(z \oplus 1)$, so $F^3 = \mathrm{id}$. Since $\mathbb{F}_{2^{2k}}^*$ is cyclic, an affine map with slope $\mu \neq 1$ has cycle type $1 + \frac{M}{\mathrm{ord}(\mu)} \times \mathrm{ord}(\mu)$ (the translation only moves the fixed point; a shorter orbit would force T^d, $d < \mathrm{ord}(\mu)$, to have a second fixed point, impossible since $\mu^d \neq 1$); the stated types therefore follow once $\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = M$ and $\mathrm{ord}(\kappa) = 2^k + 1$. Existence of such an α follows directly from Theorem 6.2. Put $q = 2^k$ and choose a primitive $\beta \in \mathbb{F}_{q^2}$ with $\beta^q \oplus \beta = 1$. Take $\alpha = \beta$. Then $1 \oplus \alpha = \alpha^q$, so

$$\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = q^2 - 1 = M, \qquad \kappa = \frac{1 \oplus \alpha}{\alpha} = \alpha^{q-1}, \qquad \mathrm{ord}(\kappa) = q + 1.$$

Here the last equality follows from $\mathrm{ord}(\alpha^{q-1}) = M/\gcd(M, q-1) = q + 1$. These are exactly the required slope orders. □

Since the K-slope has order 2^k+1, the degree over $\mathbb{F}_2$ of its minimal polynomial is $\mathrm{ord}_{2^k+1}(2) = 2k$ by Lemma 6.1; this polynomial is therefore an irreducible degree-$2k$ factor of Φ_{2^k+1}. **Remark (the τ-collapse).** Put $q = 2^k$. For $\gamma \in \mathbb{F}_{q^2}^*$ of order $q+1$, one has $\gamma^q = \gamma^{-1}$; hence $\tau = \gamma \oplus \gamma^{-1}$ satisfies $\tau^q = \tau$ and belongs to $\mathbb{F}_q$. The Frobenius identities $(1 \oplus \gamma)^{2^k-1} = \gamma^{-1}$ and $(1 \oplus \gamma)^{2^k+1} = \tau$, together with $\gcd(2^k-1, 2^k+1) = 1$, give $\mathrm{ord}(1 \oplus \gamma) = \mathrm{lcm}(\mathrm{ord}(\gamma^{-1}), \mathrm{ord}(\tau)) = \mathrm{lcm}(2^k+1, 2^k-1) = M$ whenever τ is primitive in $\mathbb{F}_{2^k}$; the pair $(u^{2^k-1}, u^{2^k+1}) = (\gamma^{-1}, \tau^{-1})$ for $u = \gamma/(1 \oplus \gamma)$ gives the same order for u. Since $\alpha = (1 \oplus \gamma)^{-1}$ and $\kappa = 1 \oplus \alpha^{-1} = \gamma$, once γ has order 2^k+1 the three order conditions on α reduce to the single condition that τ be primitive. The order assumption on γ is essential; it is not implied by τ being primitive alone. Existence of such a γ is the classical square-order statement (see [4] and [5, Theorem 1, p. 1]; for an equivalent square-order formulation, see [3, Theorem 1.5, p. 170]); it is also supplied directly by Theorem 6.2. For $q = 2^k$ the appendix proof is self-contained, so the paper does not use these classical results as external existence inputs.

The theorem is an existence statement in admissible coordinates (equivalently, for an admissible ordered generator basis). The argument of §5 applies verbatim with $q = 2^{2k}$ (taking $b \neq 0$ in the proof), so the generation $\langle T, D\rangle = \mathrm{AGL}(1, 2^{2k})$ holds for every k. For the concrete coordinate presentation of §3 the orders and cycle data were verified for $k = 1, 2, 3, 4$ (the bases are listed in the appendix): $k = 1$ (2 bits) order $12 = 4 \cdot 3$, K type $1+1 \cdot 3$, factor Φ_3 (basis $(1, 3)$: $\mathrm{Fix}(T) = \{1\}$, $\mathrm{Fix}(D) = \{2\}$, $\mathrm{Fix}(K) = \{3\}$); $k = 2$ (4 bits) order $240 = 16 \cdot 15$, K type $1 + 3 \cdot 5$, factor Φ_5; $k = 3$ (6 bits) order $4032 = 64 \cdot 63$, K type $1 + 7 \cdot 9$, factor Φ_9 (this paper); $k = 4$ (8 bits) order $65280 = 256 \cdot 255$, K type $1 + 15 \cdot 17$, linear part an irreducible degree-8 factor of Φ_{17}.

The basis is not canonical even in dimension 6: among the 720 orderings of these six generators, 572 do not even make $\mathrm{id} \oplus T$ a permutation; of the 148 that do, only 54 give T of order 63, and only 9 give the exact cycle-type triple $T = D = 1 + 63$, $K = 1 + 7 \cdot 9$. For instance $(4\,6)$ keeps $T = 1 + 63$ but changes $\mathrm{Fix}(T)$ from 38 to 11 and the cycle types of D and K to $D = 1 + 3 \cdot 21$, $K = 1 + 63$. Hence the particular entries of G and the anchors $\{38, 41, 63\}$ are coordinates of one representative, not invariants, and no uniqueness statement is made. The structural results of §5–6, however, are representative-independent: $\langle T, D\rangle = \mathrm{AGL}(1, 64)$, the third-iterate identity $F^3 = \mathrm{id}$, and the $2k$-bit existence statement hold for any admissible basis; only the numerical anchors change with the basis.

Remark (constructiveness).
Theorem 6.3 is an existence statement; its existence input is proved in Appendix B. The τ-collapse turns it into a finite search with certification. For $\tau = \gamma \oplus \gamma^{-1} \in \mathbb{F}_{2^k}$, the Artin–Schreier identity ($x^2 + \tau x + 1$ *is irreducible over* $\mathbb{F}_{2^k}$ *iff* $\mathrm{Tr}_{\mathbb{F}_{2^k}/\mathbb{F}_2}(1/\tau) = 1$) gives 2^{k-1} candidate values of τ. For a candidate, fix an explicit representation of $\mathbb{F}_{2^{2k}}$ and take a root γ of $x^2 + \tau x + 1$ (found, for instance, by exhaustive search); the collapse gives the required orders for $1 \oplus \gamma$ and $\gamma/(1 \oplus \gamma)$ precisely when τ is primitive *and* $\mathrm{ord}(\gamma) = 2^k + 1$. The second condition is not automatic from the first: for example, when $k = 6$ and $\mathbb{F}_{64} = \mathbb{F}_2[t]/(t^6 + t + 1)$, the element $\tau = t^5 + t^3 + 1$ is primitive and $\mathrm{Tr}(1/\tau) = 1$, but the roots of $x^2 + \tau x + 1$ have order 13, not 65. The theorem of Appendix B guarantees that a jointly successful candidate exists, so a deterministic enumeration that tests both conditions terminates. Certifying them requires factoring $2^k - 1$ and $2^k + 1$ (or an equivalent exact-order test). This procedure supplies a deterministic construction with certification; no factorization-free closed formula is supplied here. A necessary restatement is that $u = 1/\tau$ is a primitive element of $\mathbb{F}_{2^k}$ with $\mathrm{Tr}_{\mathbb{F}_{2^k}/\mathbb{F}_2}(u) = 1$; equivalently, its degree-k minimal polynomial has x^{k-1}-coefficient 1. This condition alone is still not sufficient for admissibility, because the root-order condition above must also be checked.

7 Remarks

The fixed point 38 is a "power-layer" origin (domain zero after relabeling), while the value 0 is the additive origin; no interpretive claim is made here. In the value set $\{0, \ldots, 63\}$ the 9-multiples are exactly the 8 self-doubled words ($x_i = x_{i+3}$), a 3-dimensional subspace. To parametrize the long orbit, define

$$\theta(j) = T^j(63), \qquad j \in \mathbb{Z}/63.$$

This is a bijection onto $V \setminus \{38\}$. Apply the CRT isomorphism to the *iteration exponent* j, sending it to ($j \bmod 7, j \bmod 9$). Thus the all-ones word $\theta(0) = 63$ corresponds to $(0, 0)$, whereas the word $0 = \theta(62)$ corresponds to $(6, 8)$, since $T(0) = 63$. Reduction of the integer word labels modulo 63 does not give this parametrization.

The cyclotomic factor Φ_{2^k+1} reflects the identity $(2^k - 1)(2^k + 1) = 2^{2k} - 1$; the single fixed point of K accounts for the remaining point. The operator's ordinary orbits on scalar affine maps admit the following complete description.

Proposition 7.1. *Let $Q = 2^{2k}$ and $\mathcal{A}_Q = \{f_{a,c} : x \mapsto ax \oplus c \mid a \in \mathbb{F}_Q \setminus \{0, 1\},\ c \in \mathbb{F}_Q\}$. The operator F acts on $\mathcal{A}_Q$ with two singleton orbits and $\big(Q(Q-2) - 2\big)/3$ orbits of length three. Its fixed maps are exactly $f_{a,0}$ with $a^2 + a + 1 = 0$. On the invariant subset $c \neq 0$, every orbit has length three.*

Proof. In coefficient pairs,

$$F(a, c) = \left((1 + a)^{-1}, \frac{c}{1 + a}\right).$$

The new slope again lies outside $\{0, 1\}$, and Lemma 4.2 gives $F^3 = \mathrm{id}$, so each orbit has length one or three. A fixed pair must satisfy $a = (1 + a)^{-1}$, equivalently $a^2 + a + 1 = 0$; the constant-term equation is $c(1 + a) = c$, so $ac = 0$ and $c = 0$. Conversely these conditions give a fixed pair. The polynomial $a^2 + a + 1$ has two distinct roots in $\mathbb{F}_Q$, since $\mathbb{F}_4 \subseteq \mathbb{F}_Q$. There are $Q(Q - 2)$ pairs in total, yielding the stated counts. Multiplication of c by $(1 + a)^{-1} \neq 0$ preserves $c \neq 0$, and no fixed pair lies in that subset. □

This proposition concerns scalar affine maps in field coordinates. Analogous cycle-type correspondences for other factorizations of $q^n - 1$ are a possible direction for further investigation.

References

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A Computational appendix

All numerical claims (fixed points, cycle types, group order 4032, and the characteristic polynomials of L_T, L_D, and L_K) were verified computationally. The script `verify_paperA_structure.py` reproduces the coordinate, cycle, group-order, basis-ordering, and τ-collapse data, as well as the $\mathbb{F}_{64}$ instance of Proposition 7.1 and the CRT exponent convention. It computes the characteristic polynomials as $\det(tI + L)$ by the permutation expansion in characteristic two. The separate `verify_k1_29.py` checks the finite PTE certificate. Both scripts are included in the companion repository. Relative to the standard basis $e_1, \ldots, e_6$, the linear part of the main map is

$$L_T = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

Formalization. The affine group-generation argument (for a field $\mathbb{K}$ with a primitive element α and $c \neq 0$, the maps T and D generate $\mathrm{AGL}(1, \mathbb{K})$, the core of §5 and §6) and the algebraic lemmas used in it (the commutator is a translation; conjugation of translations by T) have been formalized and machine-checked in Lean 4 using mathlib. An additional conditional Lean file formalizes the field-level Frobenius identities and the resulting order-theoretic τ-collapse under an explicit order interface; it does not formalize the paper proof of the existence theorem. The concrete cycle-type data of §3–4 were verified by independent computation. The existence part of Theorem 6.3 is proved in Appendix B; its analytic part is not formalized in Lean, and its finite range is discharged by the deterministic certificate there. Thus Cohen's theorem is not used as a proof dependency; the classical references are cited only for historical attribution. The Lean source is available from the author; the finite verifier is included as `verify_k1_29.py`.

Bases for the coordinate presentation

For the coordinate presentation $T(x) = (2^{2k} - 1) \oplus \mathrm{coords}_G(x)$ on $2k$ bits the following bases give the stated group order $\langle T, D \rangle$ and K-cycle type:

k	basis G	$\langle T, D \rangle$	K type	factor
1	$(1, 3)$	$12 = 4 \cdot 3$	$1 + 1 \cdot 3$	Φ_3
2	$(1, 2, 5, 13)$	$240 = 16 \cdot 15$	$1 + 3 \cdot 5$	Φ_5
3	$(63, 33, 18, 49, 1, 21)$	$4032 = 64 \cdot 63$	$1 + 7 \cdot 9$	Φ_9
4	$(134, 136, 241, 71, 34, 65, 3, 143)$	$65280 = 256 \cdot 255$	$1 + 15 \cdot 17$	a degree-8 factor of Φ_{17}

All four rows were verified by independent computation. The 720-case count was reproduced by enumerating all orderings of the six displayed vectors, solving the corresponding $\mathbb{F}_2$ linear system

for coords$_G$, and recording the cycle types of T, D, K whenever the two inverses exist.

Explicit τ-collapse data

With $\mathbb{F}_{2^{2k}} = \mathbb{F}_2[t]/(f)$ and elements written as polynomials in t, the following rows realize the order conditions of Theorem 6.3:

k	f	γ	$\tau = \gamma \oplus \gamma^{-1}$	$\alpha = (1 \oplus \gamma)^{-1}$
1	$t^2 + t + 1$	t	1	t
2	$t^4 + t + 1$	t^3	$t^2 + t + 1$	t
3	$t^6 + t + 1$	$t^2 + t$	$t^4 + t^3 + 1$	$t^5 + t^3 + t^2$
4	$t^8 + t^4 + t^3 + t^2 + 1$	$t^3 + t^2 + t + 1$	$t^7 + t^4 + t^3 + 1$	$t^6 + t^4 + t^3 + t^2 + 1$

In each row $\mathrm{ord}(\gamma) = \mathrm{ord}(1 \oplus \alpha^{-1}) = 2^k + 1$, τ is primitive in $\mathbb{F}_{2^k}$, and $\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = 2^{2k} - 1$ (all verified). The defining polynomial f is a field-representation choice and need not coincide with the characteristic polynomial p_T of (1); for instance, the $k = 3$ row uses $f = t^6 + t + 1$, while $p_T(t) = t^6 + t^5 + t^3 + t^2 + 1$.

B Self-contained proof for the even-characteristic case

Proof of Theorem 6.2. The statement is classical (see [4] and [5, Theorem 1, p. 1]; it also follows from the equivalent square-order formulation in [3, Theorem 1.5, p. 170]); we give a self-contained proof in the even-characteristic case used above. Cohen and Presern [6] also give a self-contained treatment of prescribed traces. This appendix reconstructs the classical trace-one argument and makes its finite verification explicit; no priority is claimed for the existence result or the character-sum method. Put $q = 2^k$, $L = \mathbb{F}_{q^2}$, $K_0 = \mathbb{F}_q$, and $M = q^2 - 1 = (q-1)(q+1)$. Let

$$E_1 = \{x \in L : x^q + x = 1\}.$$

The trace map $L \to K_0$ is surjective and K_0-linear, so $|E_1| = q$. If $x \in E_1$, then $x \notin K_0$ and its minimal polynomial over K_0 is $X^2 + X + xx^q$. Hence it is enough to prove that E_1 contains an element of order M.

Let ψ be the nontrivial additive character of $\mathbb{F}_2$, and put $\psi_L(x) = \psi(\mathrm{Tr}_{L/\mathbb{F}_2}(x))$ and $\psi_{K_0}(a) = \psi(\mathrm{Tr}_{K_0/\mathbb{F}_2}(a))$. For a multiplicative character χ of L^*, extend χ to 0 by $\chi(0) = 0$ and define

$$g_L(\chi) = \sum_{x \in L^*} \chi(x)\psi_L(x), \qquad g_{K_0}(\chi) = \sum_{a \in K_0^*} \chi(a)\psi_{K_0}(a), \qquad S(\chi) = \sum_{x \in E_1} \chi(x).$$

If χ is nontrivial on L^*, then $|g_L(\chi)| = q$; if $\chi|_{K_0^*}$ is nontrivial, then $|g_{K_0}(\chi)| = \sqrt{q}$. Indeed, the usual squared-modulus calculation gives

$$|g_L(\chi)|^2 = \sum_{t \in L^*} \chi(t)^{-1} \sum_{a \in L^*} \psi_L(a(1-t)) = q^2,$$

and the same argument in K_0 gives q.

We first compute $S(\chi)$. For $\alpha \in K_0$ set $E_\alpha = \{x \in L : x^q + x = \alpha\}$; these sets partition L, and $E_\alpha = \alpha E_1$ for $\alpha \neq 0$. If $\chi^{q+1} \neq 1$, then the restriction of χ to K_0^* is nontrivial. Since $E_0 = K_0$, its contribution is $\sum_{x \in E_0} \chi(x) = 0$; hence

$$g_L(\chi) = \sum_{\alpha \in K_0} \psi_{K_0}(\alpha) \sum_{x \in E_\alpha} \chi(x) = S(\chi) g_{K_0}(\chi),$$

so $|S(\chi)| = q/\sqrt{q} = \sqrt{q}$.

Suppose now that $\chi^{q+1} = 1$ and $\chi \neq 1$. Then $\chi|_{K_0^*} = 1$. Let $H = \{h \in L^* : h^{q+1} = 1\}$; since $\gcd(q-1, q+1) = 1$, the group L^* is the direct product $K_0^* \times H$. Write $x = ch$ with $c \in K_0^*$ and $h \in H$. The condition $x^q + x = 1$ becomes $c(h + h^{-1}) = 1$; thus $h \neq 1$ and $c = (h + h^{-1})^{-1}$. This gives a bijection between $H \setminus \{1\}$ and E_1, and therefore

$$S(\chi) = \sum_{h \in H,\ h \neq 1} \chi(h) = \sum_{h \in H} \chi(h) - 1 = -1,$$

because $\chi|_H$ is nontrivial.

We shall use the primitive-element indicator

$$\mathbf{1}_{\mathrm{ord}(x)=M} = \frac{\varphi(M)}{M} \sum_{\delta \mid M} \frac{\mu(\delta)}{\varphi(\delta)} \sum_{\mathrm{ord}(\chi)=\delta} \chi(x). \tag{2}$$

To see this, fix a generator g of L^* and write $x = g^a$. The left side is the indicator of $\gcd(a, M) = 1$. For a character with $\chi(g) = e^{2\pi i j/M}$ and order $\delta = M/\gcd(M, j)$, its unnormalized Fourier coefficient is the Ramanujan sum $c_M(j) = \mu(\delta)\varphi(M)/\varphi(\delta)$; Fourier inversion gives (2). (The displayed identity for $c_M(j)$ follows by Möbius inversion.)

Assume for contradiction that E_1 has no element of order M. Summing (2) over E_1 gives

$$0 = \frac{\varphi(M)}{M} \sum_{\delta \mid M} \frac{\mu(\delta)}{\varphi(\delta)} S_\delta, \qquad S_\delta = \sum_{\mathrm{ord}(\chi)=\delta} S(\chi).$$

Let $s = \omega(M)$ and $s' = \omega(q+1)$ be the numbers of distinct prime divisors. The $\delta = 1$ term is q. If $\delta \mid q+1$ and $\delta > 1$, then each character of order δ satisfies $\chi^{q+1} = 1$, so the corresponding contribution is $-\mu(\delta)$; summing over $\delta \mid q+1$, $\delta > 1$, gives 1. For the remaining squarefree divisors $\delta \nmid q+1$, the character sums are bounded by $\sqrt{q}$, and their number is at most $2^s - 2^{s'}$. Consequently,

$$q + 1 \leq (2^s - 2^{s'})\sqrt{q}. \tag{B.1}$$

Since $q + 1 > \sqrt{q}$, this implies

$$2^s > 2^{k/2}, \qquad \text{so} \qquad s > k/2.$$

In particular $k < 2s$. Let $p_1 < p_2 < \cdots$ be the primes and $P_s = p_1 \cdots p_s$. Since $M = 2^{2k} - 1$ has exactly s distinct prime divisors, $M \geq P_s$. We have

$$P_{16} = 32589158477190044730 > 2^{64}.$$

For $n \geq 16$, the ratio

$$\frac{P_{n+1}/2^{4(n+1)}}{P_n/2^{4n}} = \frac{p_{n+1}}{16}$$

is greater than 1, because $p_{n+1} \geq 59$. Hence $P_s > 2^{4s}$ for every $s \geq 16$. But $k < 2s$ gives

$$M = 2^{2k} - 1 < 2^{2k} < 2^{4s},$$

so $s \geq 16$ is impossible. Thus $s \leq 15$ and $k < 2s \leq 30$, i.e. $k \leq 29$.

It remains to cover this finite range. In the table below, m_k is an irreducible polynomial of degree k over $\mathbb{F}_2$, encoded by the binary coefficient vector of the displayed hexadecimal integer; thus $K_0 =$

$\mathbb{F}_2[x]/(m_k)$. The element $a_k \in K_0$ is encoded similarly. For each row let t be a root of X^2+X+a_k in $K_0[t]/(X^2+X+a_k)$. The deterministic verifier checks that m_k is irreducible, that $\mathrm{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$, that $t^{2^k} = t+1$, and that $t^M = 1$ while $t^{M/p} \neq 1$ for every prime $p \mid M$. The last two checks show that $\mathrm{ord}(t) = M$; the third shows that $\mathrm{Tr}_{L/K_0}(t) = 1$. A verifier for the table, `verify_k1_29.py`, is included as an ancillary file; its SHA-256 hash is `042CE7F74EDFF13ADE5C6285E42CB3D15491FC25C65B99BC670D256047EE8D9F`.

k	m_k	a_k
1	0x3	0x1
2	0x7	0x3
3	0xb	0x5
4	0x13	0xd
5	0x25	0x13
6	0x43	0x34
7	0x83	0x47
8	0x11b	0xc8
9	0x203	0x2b
10	0x409	0xcb
11	0x805	0x525
12	0x1009	0x39e
13	0x201b	0x1ec4
14	0x4021	0x3717
15	0x8003	0x62bf
16	0x1002b	0x1e6e
17	0x20009	0x1155b
18	0x40009	0x2ee84
19	0x80027	0xb9af
20	0x100009	0x6aee2
21	0x200005	0x15750b
22	0x400003	0x24a150
23	0x800021	0x6c92f9
24	0x100001b	0x389ec4
25	0x2000009	0x6801ad
26	0x400001b	0x2707d2e
27	0x8000027	0x6e9ab1e
28	0x10000003	0xfac7cd2
29	0x20000005	0xc69b55e

The table therefore supplies the required element for every $k \leq 29$. Combining this with the contradiction for $k \geq 30$ proves the theorem. $\square$